

Rule-Based Matching for Real Estate Features Detection

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Abstract. Most of the information about real estate for sale in the Buenos Aires province, Argentina is unstructured, which means that it does not always follow the same format, making extraction a challenging process. Variability in wording, human errors, noise, and incomplete data further complicate the task. Given the large volume of information available, automated techniques are required to transform unstructured text into structured data. This article presents an approach to extract attribute-value pairs from the information contained in the property listings for the province of Buenos Aires, in order to incorporate this data into a knowledge graph. The approach uses pattern-based information extraction for 17 features with an exhaustive evaluation over two datasets: a ground truth labeled by experts and a dataset containing a real-world use case. The results demonstrate accurate values.

Keywords: Information Extraction · Rule-based matching · Natural Language Processing · Knowledge Graph Completion

1 Introduction

Most of the information available on the Web is published in natural language, making it challenging for machines to automatically process those data and draw inferences from them. Automatic techniques are required to convert text into information. To address this challenge, information extraction (IE) [3] techniques are used to automatically identify and extract structured data from unstructured sources.

Several cases, the result of the information extraction is inserted in a knowledge graph (KG)[9]. Knowledge graphs [5] are structured representations of information in the form of entities connected by relationships. Since knowledge graphs support automatic reasoning[14,13], they can be highly effective in improving search engines and other analytical tools.

Specifically, the information related to the Real Estate market is more available and updated on the Web rather than official information. Real estate market information appears on the Web on specific sites, including a typical advertising description with a set of structured information. The description is written in natural language including information related to different relevant attributes in the description of the building. This work is focused in real estate listing published in Buenos Aires, Argentina which involves the use of Spanish language, which is not well developed in most of the NLP tools. Hence, the difficult of attributes detection in natural language is combined with the specific language making that detecting this type of attribute is currently a challenge.

Thus, the addressed problem is to extract attribute-value pairs embedded in the descriptions of real estate listings to improve a knowledge graph of real estate domain. An attribute-value pair describes a relationship between a feature (attribute) and its corresponding information (value). For example, an attribute value pair could be {"address" : "calle 120 y 50"}.

There are several approaches to information extraction using Natural Language Processing (NLP). A commonly used method is Rule-Based Matching (RBM), which has been applied in various contexts, including medical record analysis [4, 7]. This approach enables the identification of specific entities through predefined rules, facilitating the structuring of information extracted from unstructured documents. However, rule-based methods are not the only alternative for NLP. Supervised learning techniques, such as Named Entity Recognition (NER), have proven effective in automatically identifying key terms in texts [8, 12]. In the domain of medical document information extraction, studies like [6] have analyzed the current state of these techniques and potential future directions. Another relevant approach is the use of deep learning models. In the case of evidence-based dietary recommendations, a rule-based named-entity recognition method has been employed for knowledge extraction, as described in [2] a rule-based named-entity recognition method for knowledge extraction of evidence-based dietary recommendations. For the specific case of real estate listings, few works are published. For example, a dissertation thesis about detecting distressed real estate using NLP [10] with a english listing analysis. In Spanish, Tanevitch et. al [11] evaluate different IE approaches to extract features in real-estate descriptions.

This article presents an approach for detecting real estate features in natural language descriptions of real estate listings. The approach uses pattern-based information extraction for 17 features. In addition, this article performs an exhaustive evaluation using traditional performance metrics such as precision and recall over two datasets: a ground truth labeled by experts and a dataset containing a real-world use case. Finally, the proposed approach is compared with the one proposed by Tanevitch et. al. [11] on the same dataset, demonstrating an overall improvement in performance.

The organization of this article is detailed as follows. Section 2 introduces an overview of the workflow from getting raw data to make it structured. Next, section 3 describes the real estate features that should be extracted from the

listing description. Then, section 4 has a simple explanation of the approach used for natural language processing. This is followed by the section 5, which details the process of extracting attribute-value pairs for some of the most relevant attributes. Moving on, the section 6 explains the data and metrics used to evaluate the created evaluation patterns. Section 7 highlights the metrics of the information collected. Finally, section 8 indicates conclusions and possible improvements to be applied to the project as further work.

2 Data Workflow Overview

The process of obtaining and structuring data for the real estate observatory involves to get the data and to structure it according a semantic model. Figure 2 shows a data workflow overview starting in obtaining data from the Web and to save it on a Knowledge Graph. The Knowledge Graph follows a semantic formalization in terms of an ontology. A real-estate-domain ontology [1] defines the real-estate related components (such as Real Estate), their attributes (like the address) and properties (i.a. publishedIn).

As shown in Figure 2, when data is tabular it can be directly mapped to the Knowledge graph [1]. However, in natural language descriptions, since features are embedded in the text, it is needed to apply an extra step to extract the attribute-value pairs list, and then mapping them to the KG. The contribution of this article expands on this module.

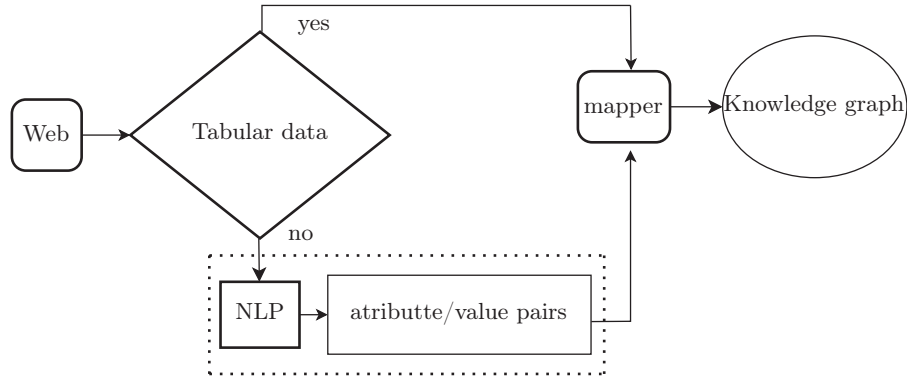


Fig. 1. Data Workflow Image

3 Real Estate Features Description

This section describes the main characteristics of the attributes to be detected. As real estate listings are published in the province of Buenos Aires, Argentina,

the language in which all of this are written is in Spanish. However, the following section will detail the features in English and some cases are detailed between parenthesis with the Spanish version.

In particular, the real estate listing descriptions that are analyzed are written for different audiences. Some listings are intended for those who are going to buy a single family house, while other publications use a different vocabulary, as they are intended for more technical profiles.

Address is one of the most relevant attributes to be detected since it allows spatial localization. Four main writing formats can be generalized: mentions of a street, with the height of the lot ("Av. 19 n° 134"), mentions of 2 intersecting streets (sp "calle San Martín y Villegas"), mentions of the main street and between which it is located (sp "Calle 1 entre 2 y Mitre"), and addresses based on lots (sp "lote 3 manzana E") [11]. There are other references to less precise locations that are not considered in this analysis, for example "Route 36 KM 11".

The **FOT** (in Spanish *Factor de Ocupación Total*) is a coefficient that determines the maximum buildable area of a parcel or lot. There are different types of FOT such as residential, commercial, industrial and others. In some cases the type of FOT may include awards for more building. Finally, the units of the FOT can be in square meters and its variants or in a proportion of the surface of the lot. In the advertisements it usually appears in two forms, one where it is indicated with the word "fot" or "FOT" followed by the numerical value (for example "fot 0.9") or also in natural language of the style "FOT maximum residential of 3" (sp. "FOT máximo residencial de 3").

It is necessary to identify whether the lot occupied by the property is **irregular** or not. The attribute seeks to identify land with irregular dimensions and determine a truth value. Generally, the listings includes the word "irregular", as well as words such as "hammer", "trapezoid" or "triangular" (sp. "martillo", "trapezio", "triangular") in reference to the shapes that the lot possesses.

The lot side **dimensions** are the front and back measurements of a lot. When lot is squared, listings usually refer to this feature (for example, sp "10 x 50", "Frente: 10, Fondo: 50"), and may include the unit of measure. In irregular lots more sizes can be mentioned since the shape is not squared, for example "10 x 50 x 30 x 40".

Some lots are located at the intersection of two streets, meaning they are on a **corner**. The traditional ways in which this attribute appears are with the words "corner", or phrases such as "it is located in the corner". The challenges for this attribute is the appearance of references to nearby spaces, such as "police station on the corner".

The characteristic **neighborhood** makes it possible to determine in which subdivision of the locality the property is located. This characteristic is difficult to recognize, even for humans, because the word "neighborhood" is usually omitted in the description and the property is described as being within a spatial delimitation with the expression "located in".

Hence, it is also important to identify properties located in a **gated urbanization**. These are usually identified by phrases such as "closed urbanization" or "closed neighborhood". In addition, some advertisement include additional information about other types of amenities, such as "club house". The properties may belong to semi-enclosed neighborhoods or **semi-gated urbanizations**. This feature is similar to gated community, except that it includes the term "semi-closed" or similar.

The **fronts** attribute indicates the number of sides the lot has that are adjacent to any of the surrounding streets. Generally they have a single facade, but there are lots that have access from more than one street, and in those cases they will have more than one facade. The most common formats to mention the facades of a lot are "lot with 2 facades", "exits to 3 streets", "the lot has access from 3 fronts", or "lot with triple facade".

To indicate that the property has a **swimming pool**, the word "swimming-pool" is usually used. The difficulty arises when multiple listings mention the community swimming pools of a gated community or condominium. This feature usually appears with other condominium amenities such as a gym, tennis court or multipurpose room. And another less common difficulty, but one that also occurs, is the mention of canvas swimming pools, which are not considered in the detection of the feature.

The **legal status** about the possessory right of the property is analyzed to determine whether the bidder has the right to the property or not. In particular, the cases in which the legal condition of the land is of the type of usucapion, cession of possession rights or possesses a contract of sale are detected. These conditions are usually expressed with phrases containing the words "title", "rights", "usucapion" or phrases such as "does not pay expenses until possession".

If the building referred to has not yet been built, but the project has been approved, the offer is called a **pre-sale**. The word "pre-sale" usually appears in these cases, but it is also used to refer to a medium or long term investment, such as financing, trust, installment or medium term possession. Phrases such as "development opportunity" (sp. *"posibilidad de desarrollo"*), or "building proposal" (sp. *"propuesta de construcción"*) are some examples.

The lots that cannot be subdivided are labeled under the **undivided** part regime. In the listings is mentioned that the lot is a fraction of a larger lot without legal subdivision, has an undivided part, says "undivided fraction" or says "it has an undivided deed" (sp. *"posee escritura indivisa"*). In some cases, however, the description states that they are not undivided or that the "subdivision is in process," making the task of identification more difficult.

Identifying whether the lot has a part of its structure **to demolish** is relevant. This is indicated in those advertisements where it is stated that the property has a structure that does not add value, either because it is to be repaired or destroyed, or in those ads where the structure does not add value to the sale. Terms such as "to demolish" (sp. *"a demoler"*) or "ideal for builder" (sp. *"ideal constructor"*) are used.

The **multioffer** is an attribute that identifies those advertisements in which several lots are offered in a single description. In general, they are characterized by the use of the plural in the term lots, a number that indicates several lots for sale. For example, "5 lots for sale" or "10m2 lots".

Those publications that describe an improvement that increases the value of the land, which can be due to a house, a gate, a fence, a wall, etc., are considered to meet the attribute called **monetizable**. This feature is complex because we have to make sure that this construction corresponds to the land and that it is not an intermediate project.

The **condominium lot** attribute attempts to identify the parcels that are subject to the horizontal property regime. One of the difficulties with this feature is that several notices include a denial, indicating that the lot is not subject to horizontal property.

4 Information Extraction Rule Based

Rule-based matching is a technique for extracting information from unstructured text by using predefined rules. The advantages of this method are that it is easy to understand the rules by which information is extracted, not much data is needed and the analysis does not require much computation. In general, the disadvantages are usually related to the large amount of work that must be applied to define the patterns for a given context, it is not a good approach for identifying valuable information when faced with new formats in which the information is found and it is necessary to refine the constructed patterns, so the tool is usually run several times, correcting the patterns to improve performance. One application of this technique could be done by manually defining unique patterns based on explicit syntactic rules to identify words, phrases, or syntactic structures. In particular, it is recommended to use this methodology to deal with features where we have few examples, and for well-defined and constant patterns, such as date recognition, URL formats, etc.

To carry out this strategy, we use Python's SpaCy library, which consists of three rule-based matching engines that will be explained below. The first one we will explain is the *Matcher*, which is able to receive sequential patterns to search within a description, asking for nouns, adjectives, prepositions, etc. in a specific order. Another tool is the *PhraseMatcher*, which only requests text in a specific order. Finally, the library provides the *DependencyMatcher*, which is able to find syntactic dependencies between words that do not have to be consecutive. This tool is very good for cases where we want to find patterns that do not follow a fixed order.

5 Matchers Description

This section describes the most interesting matchers, however the complete implementation of all the matchers is available at <https://github.com/cientopolis/OVS-extractor-idis>.

Regarding the detection of the address attribute, there is a Matcher structured in the four possible formats, each of which has its own set of patterns.

In order to introduce the address strategy, the following matcher example will detail one of the four cases. In particular, when an address that mentions a street between two others. In detail, the particular case that contains the names of three streets, one of which is numeric followed by the house number, for example *sp "calle 7 n° 1231, entre calle Moreno y San Martin bis"*.

One extract of the matcher is described in Listing 1.1. It shows from line 9 to 11 the detection of the street name, then lines 12 to 15 the house number. Following, lines 16 to 17 a connector, then line 18 to 20 the 18-20 the name of the first adjacent street, line 21 for a connector, and finally lines 22 to 24 detect the name of the second adjacent street.

Listing 1.1. Extract from the matcher to address detection

```

1 CALLE_SINONIMOS = ["bv.", "bv", "avs", "avs.", "av", "av", "...]
2 CALLE_SEGMENTO = ["bis", "Bis", "BIS"] + LETRA_MAYUSCULA
3 NUMERO_SINONIMOS = ["numero", "numeros", "nro", ... ]
4 ANTE_NUMERO = ["km", "al", "altura", "altura:", "alt", ... ]
5 MEDIDAS = ["metro", "metros", "m", "ms", "mt", "mts", ...]
6 ENTRE = ["e/", "entre", "a", "a/", "esquina", "esq", "esq."]
7 INTERSECCION = ["y", "e", "esquina", "esq", "esq."] ...
8 [
9     {"LOWER": {"IN": CALLE_SINONIMOS}, "OP": "?"},
10    {"POS": "NUM"},
11    {"ORTH": {"IN": CALLE_SEGMENTO}, "OP": "?"},
12    {"LOWER": {"IN": NUMERO_SINONIMOS+ANTE_NUMERO}, "OP": "*"},
13    {"IS_PUNCT": True, "OP": "?"},
14    {"LIKE_NUM": True, "OP": "?"},
15    {"IS_PUNCT": True, "OP": "?"},
16    {"LOWER": {"IN": ENTRE}},
17    {"POS": "DET", "OP": "?"},
18    {"LOWER": {"IN": CALLE_SINONIMOS}, "OP": "?"},
19    {"POS": {"IN": ["PROPN", "NUM"]}, "OP": "{1,3}"},
20    {"ORTH": {"IN": CALLE_SEGMENTO}, "OP": "?"},
21    {"LOWER": {"IN": INTERSECCION}},
22    {"LOWER": {"IN": CALLE_SINONIMOS}, "OP": "?"},
23    {"POS": {"IN": ["PROPN", "NUM"]}, "OP": "{1,3}"},
24    {"ORTH": {"IN": CALLE_SEGMENTO}, "OP": "?"}
25 ] ...

```

The following is an example of a pattern for the address format as a lot, because this is completely different from the other three formats. In this case, it would be matched with something similar to *sp "lote nro. 5 en manzana E"*. The fragment of the matcher can be visualized in the listing 1.2. Lines 7 to 9 detect the lot name, line 10 for a connector, then 11 to 13 detect the block.

Listing 1.2. Extract from the matcher to lot address detection

```

1 ...
2 NOMBRE_LOTE = ["NUM", "PROPN"]

```

```

3 SOBRE_SINONIMOS = ["en"]
4 MANZANA_SINONIMOS = ["manzana", "mz", "mz.", "mza", "mza."]
5 LETRA_MAYUSCULA = [ ... ] ...
6 [
7     {"LOWER": "lote"},
8     {"LOWER": {"IN": NUMERO_SINONIMOS}, "OP": "*"},
9     {"POS": {"IN": NOMBRE_LOTE}},
10    {"LOWER": {"IN": SOBRE_SINONIMOS}, "OP": "?"},
11    {"LOWER": {"IN": MANZANA_SINONIMOS}},
12    {"LOWER": {"IN": NUMERO_SINONIMOS}, "OP": "*"},
13    {"TEXT": {"IN": LETRA_MAYUSCULA}}
14 ] ...

```

Six patterns have been created for the FOT matcher, combining the different alternatives for its appearance. The listing 1.3 shows a fragment of the matcher for cases of the form sp. *"fot residencial de 50 m2"*. Line 6 detects the word "fot" and its variants, then line 7 detects the type of fot, line 8 detects connectors, line 9 detects the fot number, and line 10 detects unity.

Listing 1.3. Extract from the matcher to detect FOT

```

1 ...
2 TIPOS_FOT = ["residencial", "comercial", "industrial", ...]
3 FOT_SINONIMOS = ['fot', 'f.o.t', 'F.O.T', 'Fot', 'F.o.t', 'FOT:']
4 FOT_CONECTOR = ["de", ":", "es", ",", ".", "="]
5 FOT_UNIDAD = ["m2", "mts2", "metros2", "metros_cuadrados", ...] ...
6 [ {"LOWER": {"IN": FOT_SINONIMOS}},
7   {"LOWER": {"IN": TIPOS_FOT}, "OP": "?"},
8   {"TEXT": {"IN": FOT_CONECTOR}, "OP": "*"},
9   {"POS": "NUM"},
10  {"LOWER": {"IN": FOT_UNIDAD}, "OP": "?"}
11 ] ...

```

To detect the value of the fronts attribute, rules are defined based on a DependencyMatcher, which has a pattern for each of the mentioned formats. The listing 1.4 shows the pattern. The matcher includes three blocks. The first block (lines 2 to 3) detects the word "exit" (sp. *"salida"*). The second block (line 4 to 7) detects the any word with an oblique nominal syntactic dependency with the first block. And the third block, finds a word with a numerical dependency with the second block. In conclusion, the extract of Listing 1.4 could detect texts similar to "exit to 3 streets" (sp. *"salida a 3 calles"*), where *"salida"* is the root, *"calles"* has the oblique dependency to the root, and "3" has the numerical dependency to *"calles"*.

Listing 1.4. Extract from the matcher to detect fronts

```

1 [
2     { "RIGHT_ID": "frentes",
3       "RIGHT_ATTRS": {"LOWER": "salida"}},
4     { "LEFT_ID": "frentes",
5       "REL_OP": ">",

```



```

6      "RIGHT_ID": "calles",
7      "RIGHT_ATTRS": {"DEP": "obl"},},
8  {   "LEFT_ID": "calles",
9      "REL_OP": ">",
10     "RIGHT_ID": "num",
11     "RIGHT_ATTRS": {"DEP": "nummod"},},
12 ],

```

6 Evaluation

This work evaluates the performance of the rule-based approach in detecting attribute-value pairs in real estate descriptions.

The evaluation of the approach will include the analysis of the performance in the detection of attributes in terms of precision, recall and f1-score, then a comparison with previous works, and finally the performance in real dataset of real estate advertisements. The patterns are applied to a massive dataset, and to measure the results in this context based on a portion of the graph. Here, the goal is to analyze what happens with unreviewed data.

6.1 Metrics

In order to analyze the performance of the approach, precision, recall, f1-score metrics and accuracy are used.

Precision is calculated using Equation 1, where TP represents the pairs correctly recognized, and FP refers to cases where the model identified attribute-value pairs that were not actually present.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (1)$$

Precision allows us to assess the model's ability to correctly identify attribute-value pairs.

Recall measures the model's ability to identify all correct pairs and is computed according to Equation 2, where TP represents the pairs correctly recognized by the tool, and FN refers to mentions that should have been recognized but were not.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (2)$$

The f1-score is a harmonic measure that combines precision and recall. calculated using Equation 3 and is

$$\text{F1-score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3)$$

Finally, accuracy is the amount of correct elements detected overall. This is particularly useful in the evaluation of the real-case OVS dataset since the

records were randomly selected and there are features that may contain only true negative occurrences, so this metric would be the only representative one in this cases. The formula is given in the equation ??.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (4)$$

6.2 Datasets

Ground truth The ground truth is comma separated files with 321 tuples. Each tuple contain the actual listing text of a real estate advertising and the values of the 17 attributes for each description. The dataset was manually tagged by experts in the study of real state.

Real-case OVS The real case dataset is a comma separated file that contains 320,000 real estate advertising descriptions without any type of tabular data. This file represents a real-world case in the domain, as it was constructed using automated techniques. As a result, the information may be incomplete or unnormalized, and variable distribution could be uneven.

7 Results

In order to evaluate the approach, two evaluation steps were done. The first analyze the approach with the ground truth dataset and also compares the resulting performance with the one introduced by Tanevitch et al.[11]. The second evaluation uses the Real case OVS dataset, taking as a first step the occurrences of each variable across the 320,000 tuples, and then extracting 100 random tuples to rerun the approach. A manual analysis of the results was then performed for each attribute to calculate the various metrics.

7.1 Ground truth results

Table 1 resumes the values obtained for each metric comparing the performance of this article contribution with Tanevitch et al.[11] approach. The first column details the analyzed attribute. Then, there are two blocks of columns. Each block includes the values of metrics precision, recall and F1-Score. Note that in the Tanevitch et al. approach, patterns were designed for only 8 of the 17 features in question, which is why the metrics are "n/a" in several cases.

Almost all metrics were improved in the approach proposed in this contribution. And, none of the metrics decreased the value. The most significantly improvements were related to attributes address, fronts and dimensions. Address precision improved 0.75 points, recall improved 0.13 points, and consequently f1-score improved 0.38 points. Fronts precision improved 0.07 points, recall improved 0.52 and F1-score improved 0.4. And, dimensions precision improved 0.02, recall improved improved 0.26 points and F1-score 0.17. Most of

the improvements are related to the extensive of the cases in the definition of the matchers. For example, some matchers, as address extends the number of cases that Tanevitch et al. included in their approach.

Table 1. Metrics comparison: Tanevitch et al. vs actual contribution

Attribute	Tanevitch et al.			Actual contribution		
	Precision	Recall	F1 Score	Precision	Recall	F1 Score
Address	0.24	0.59	0.34	0.78	0.72	0.72
FOT	0.85	0.92	0.89	1.00	0.96	0.98
Irregular	1.00	0.70	0.83	1.00	0.90	0.94
Dimensions	0.85	0.54	0.66	0.87	0.80	0.83
Corner	0.90	0.92	0.91	0.90	0.92	0.91
Neighborhood	0.45	0.26	0.33	0.54	0.43	0.48
Fronts	0.88	0.38	0.53	0.95	0.90	0.93
Swimming Pool	0.76	0.97	0.85	0.82	0.97	0.89
Gated Urbanization	n/a	n/a	n/a	0.97	0.97	0.97
Legal Status	n/a	n/a	n/a	0.94	1.00	0.97
Semi-gated Urbanization	n/a	n/a	n/a	1.00	0.94	0.97
Pre-sale	n/a	n/a	n/a	0.94	0.94	0.94
Undivided	n/a	n/a	n/a	0.88	1.00	0.94
To Demolish	n/a	n/a	n/a	0.97	0.97	0.97
Multioffer	n/a	n/a	n/a	0.79	0.97	0.87
Monetizable	n/a	n/a	n/a	0.98	0.96	0.97

7.2 Real-estate OVS results

As a first step, the proposed approach was executed on the 320,000 tuples of the complete dataset. The goal of this execution is to count the occurrences and then, analyze in a portion of the cases the accuracy of the approach. Table 2, has two columns, the first column has the attributes and, the second column the number of detected occurrences of each attribute. As can be seen, the attribute monetizable had the most number of occurrences of 275,510 occurrences. Then, it decrease the amount of occurrences with the attributes swimming-pool, dimensions and address. Swimming-pool had 68,144 occurrences. Dimensions had 64,911 occurrences, address had 67,457 occurrences. For the rest of the attributes, the value decrease and the less appearance was for legal status with only 270 occurrences.

Second, 100 random cases present in the Real-estate OVS dataset were selected for a manual analysis in order to define precision, recall, f1-score and accuracy. The manual process analyze each description in natural language and check if the attribute appears and how the approach of this article detect or note the value of the attribute.

The results are shown in table 3. The first column contains the attributes, where the 17 defined attributes are mentioned, followed by the metrics to be used. It is important to note that some features may not be covered in the

Table 2. Number of matches detected on real-case ovs dataset

Attribute	Occurrences	Attribute	Occurrences
Address	67457	Gated Urbanization	19450
FOT	5728	Legal Status	270
Irregular	2159	Semi-gated Urbanization	1152
Dimensions	64911	Pre-sale	738
Corner	18905	Undivided	395
Neighborhood	35437	To Demolish	25857
Fronts	2350	Multioffer	3318
Swimming Pool	68144	Monetizable	275510
Condominium Lot	339		

ground truth due to their rarity in the dataset, and for this reason the model fails to detect them, obtaining an f1-score of 0. However, this is not a problem, as the accuracy metric shows that the model works correctly, confirming that it has adequately not identified these features.

The results shown that corner, dimensions and monetizable are the best recognized attributes. Corner had 0.93 points of f1-score, dimensions had 0.88 points of f1-score and monetizable 0.81 points of f1-score. Several attributes are not recognized. However, none of them included false positives. Irregular had the maximum precision value but a low recall value. That means that the matcher is not taking into account several other cases about that attribute.

The results shown that the first 8 attributes did not worsen much their f1-score with respect to the table 1, since some attributes improved (fronts and neighborhood), others remained practically the same (corner and dimensions) and others worsened (address, irregular, swimming pool). Due to their low frequency, the attributes FOT, condominium lot, semi-gated urbanization, legal status, pre-sale, undivided and to demolish were left with non occurrence (n.o.) value.

It should be noted that the multioffer and monetizable matchers acquired good metrics, even if the monetizable attribute is the one that usually has the most occurrences.

Table 3. Comparison of Precision, Recall, F1-score, and Accuracy for 17 attributes. Those attributes without occurrences are labeled with n.o.

Attribute	Precision	Recall	F1	Accuracy	Attribute	Precision	Recall	F1	Accuracy
Address	0.56	0.41	0.47	0.69	Gated Urbanization	0.60	1.00	0.75	0.98
FOT	n.o.	n.o.	n.o.	1.00	Semi-gated Urbanization	n.o.	n.o.	n.o.	1.00
Irregular	1.00	0.33	0.50	0.98	Legal Status	n.o.	n.o.	n.o.	0.99
Dimensions	0.91	0.85	0.88	0.91	Pre-sale	n.o.	n.o.	n.o.	1.00
Corner	0.87	1.00	0.93	0.98	Undivided	n.o.	n.o.	n.o.	1.00
Neighborhood	0.41	0.50	0.45	0.68	To Demolish	n.o.	n.o.	n.o.	0.97
Fronts	1.00	1.00	1.00	1.00	Multioffer	0.67	0.55	0.60	0.92
Swimming Pool	0.67	1.00	0.80	0.99	Monetizable	0.74	0.90	0.81	0.84
Condominium Lot	n.o.	n.o.	n.o.	1.00					

8 Conclusions and future work

This article introduces a rule-based matching approach to detect 17 real estate features. The article details an overview of the general workflow of a data analysis in the Land Value Observatory (OVS in spanish), then a feature explanation is described and it is followed by a detailed explanation of the most relevant implemented matchers.

Compared with a previous approach, the one introduced in this article shown a relevant improvement of the precision, recall and F1 metrics in all the attributes. Also, our approach increased the number of attributes detected with good accuracy. In addition, our approach was executed to analyse a real estate offer dataset of 320,000 advertisements from Buenos Aires, Argentina. The evaluation demonstrated several occurrences of all the attributes distinguishing monetizable and address as the most relevant, and legal status as the lowest presence.

As further work, an extended evaluation is mandatory, in order to have more evidence about the attributes that in this article appear without occurrences. In addition, the combination of this approach with other NLP techniques to improve the different performances of the matchers could be done.

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